

ARAB YOUTH SPACE HACKATHON 2026 • CHALLENGE 813 • TEAM T0049

QAYDH القيث

Heat-risk intelligence for fast-growing Gulf cities

انتصار الحبسي • نورة الهاجري • مريم البني

WHY QAYDH

A heat map says where. A city needs who, why and what next.

CHALLENGE

Summer surfaces pass 50 °C. Cooling budgets are spent case by case.

SIGNIFICANCE

People still walk to prayer, wait for buses, deliver and build outdoors at noon.

IMPACT

Shade, cool roofs and trees go first where people are exposed.

SOLUTION

Hyperspectral + thermal + people data → one ranked action map.

THE PEOPLE IN THE HEAT

11:00–20:00: above 40 °C on most summer days

Worshippers

walking to Dhuhr and Asr

Bus riders

waiting at unshaded stops

Delivery riders

idling outside restaurants

Workers

on construction sites

157,076

residents in the hottest fifth of east Riyadh

0.00 vs 0.63

mapped bus stops per 10,000 people, hottest vs rest

8.3 h

a day at or above 40 °C

Five questions, one map

WHERE

**Persistent
hotspots**

Landsat LST, 3 summers

WHEN

Danger hours

ERA5 hourly weather

WHY

**Roof · road ·
sand**

Planet Tanager 426 bands

WHO

**People
outdoors**

WorldPop +
OpenStreetMap

WHAT NEXT

**Named action
+ °C**

300 m priority cells

DATA & TOOLS

Open data.
One notebook.
Run all.

DATA

Planet Tanager hyperspectral · Landsat 8/9 SR + thermal · ESA WorldCover · Impact Observatory · WorldPop 2025 · OpenStreetMap · Microsoft building footprints · Open-Meteo/ERA5

TOOLS

Python · Planetary Computer STAC · scikit-learn · Leaflet dashboard · gIQ-ready GeoJSON

STEPS

Cloud & quality masks → automatic labels → model training → spatial validation → ranked actions → dashboard

01 · WHERE

The city grew 28% in eleven years

Built-up land rose from 167 to 214 km² (2014→2025). Magenta = 33 km² of confidently new districts.

0.890.58

built-up F1starter NDBI rule

02 · WHERE

Where it burns at 10:40 am

Median summer surface temperature. Established districts average 52.2 °C. The same cells stay hottest year after year (2024↔2025 r = 0.90).

415-0.01

cells hot 3 summersstarter NDBI ↔ heat r

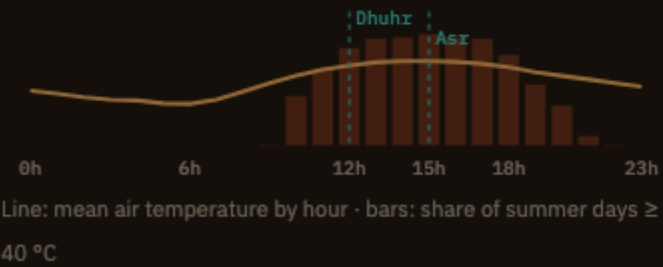
03 · WHEN

Danger runs 11:00–20:00

Air passes 40 °C on most summer days from 11:00. That covers Dhuhr and Asr prayers, school runs and lunch-rush deliveries. Peak at 15:00.

8.312

hours/day ≥ 40 °Cdays ≥ 45 °C

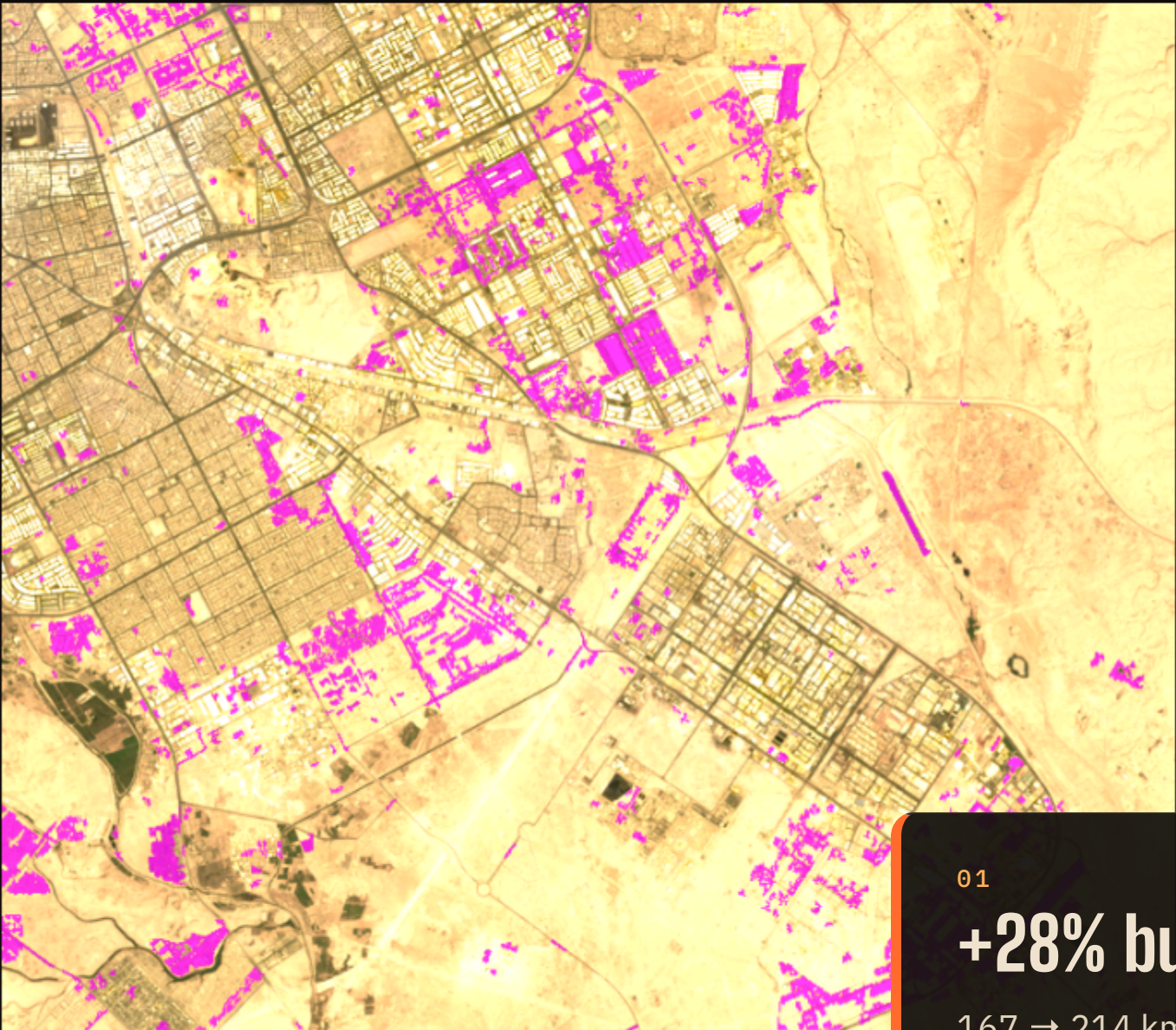


04 · WHY

WHERE

+

-



New urban 2014–2025

East Riyadh

As Sali · Saudi Arabia

WHERE

Persistent summer hotspots at 300 m

WHEN

Hours above 40 °C, from weather reanalysis

WHY

Roof, road, sand or missing greenery, read from hyperspectral spectra

WHO

Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children

WHAT NEXT

One named action per hotspot, with estimated °C and confidence

Open the last chapter, “Where to act first”, and pick a hotspot.

01

+28% built-up since 2014

167 → 214 km². Our map scores F1 0.89 against the starter rule's 0.58.

02 · WHERE

Where it burns at 10:40 am

Median summer surface temperature.

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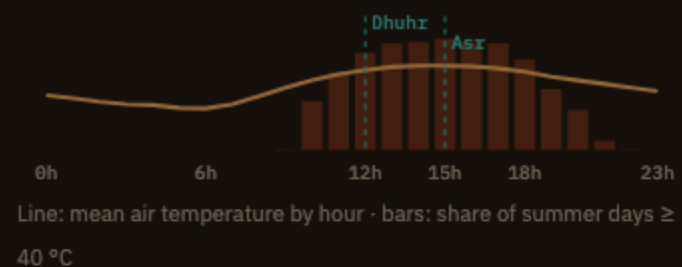
Air passes 40 °C on most summer days from 11:00. That covers Dhuhr and Asr prayers, school runs and lunch-rush deliveries. Peak at 15:00.

8.3

hours/day ≥ 40 °C

12

days ≥ 45 °C



04 · WHY

Roof, road or sand? Tanager reads the surface

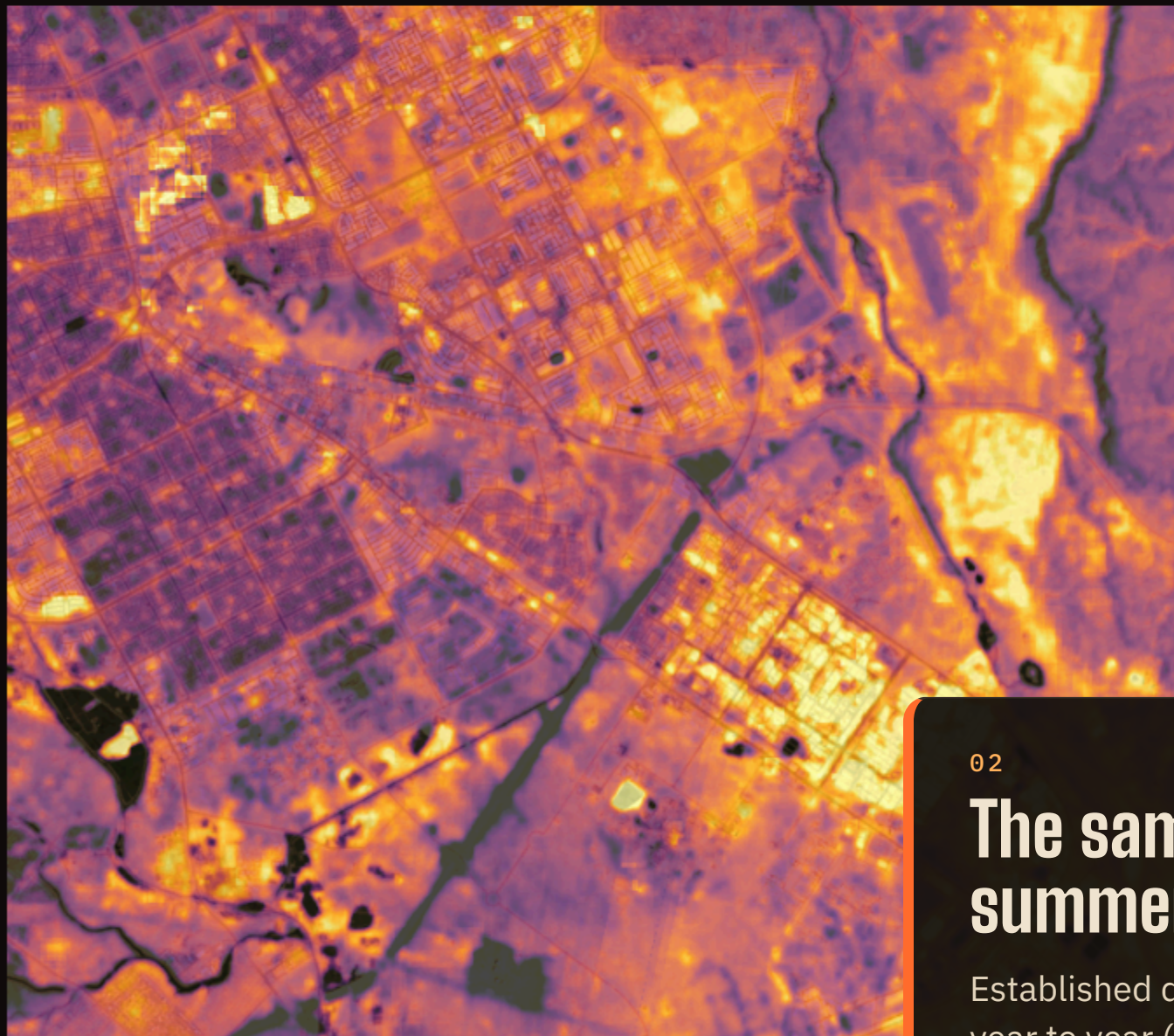
426-band Tanager spectra, labelled automatically from OpenStreetMap + Microsoft building outlines and WorldCover, sort every pixel into road, roof, vegetation and sand. Tested on tiles the model never saw.

0.76

0.75

0.38

WHERE



Surface temperature



East Riyadh

As Sali · Saudi Arabia

WHERE	Persistent summer hotspots at 300 m
WHEN	Hours above 40 °C, from weather reanalysis
WHY	Roof, road, sand or missing greenery, read from hyperspectral spectra
WHO	Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children
WHAT NEXT	One named action per hotspot, with estimated °C and confidence

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02

The same streets burn every summer

Established districts average 52.2 °C. Hot cells repeat year to year ($r = 0.90$). The starter NDBI proxy explains none of it ($r = -0.01$).

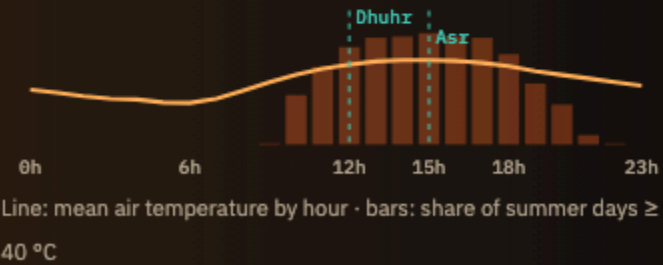
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0.760.750.38

full spectrum F16 broad bandsstarter index rules

05 · WHO

Who is outside in it

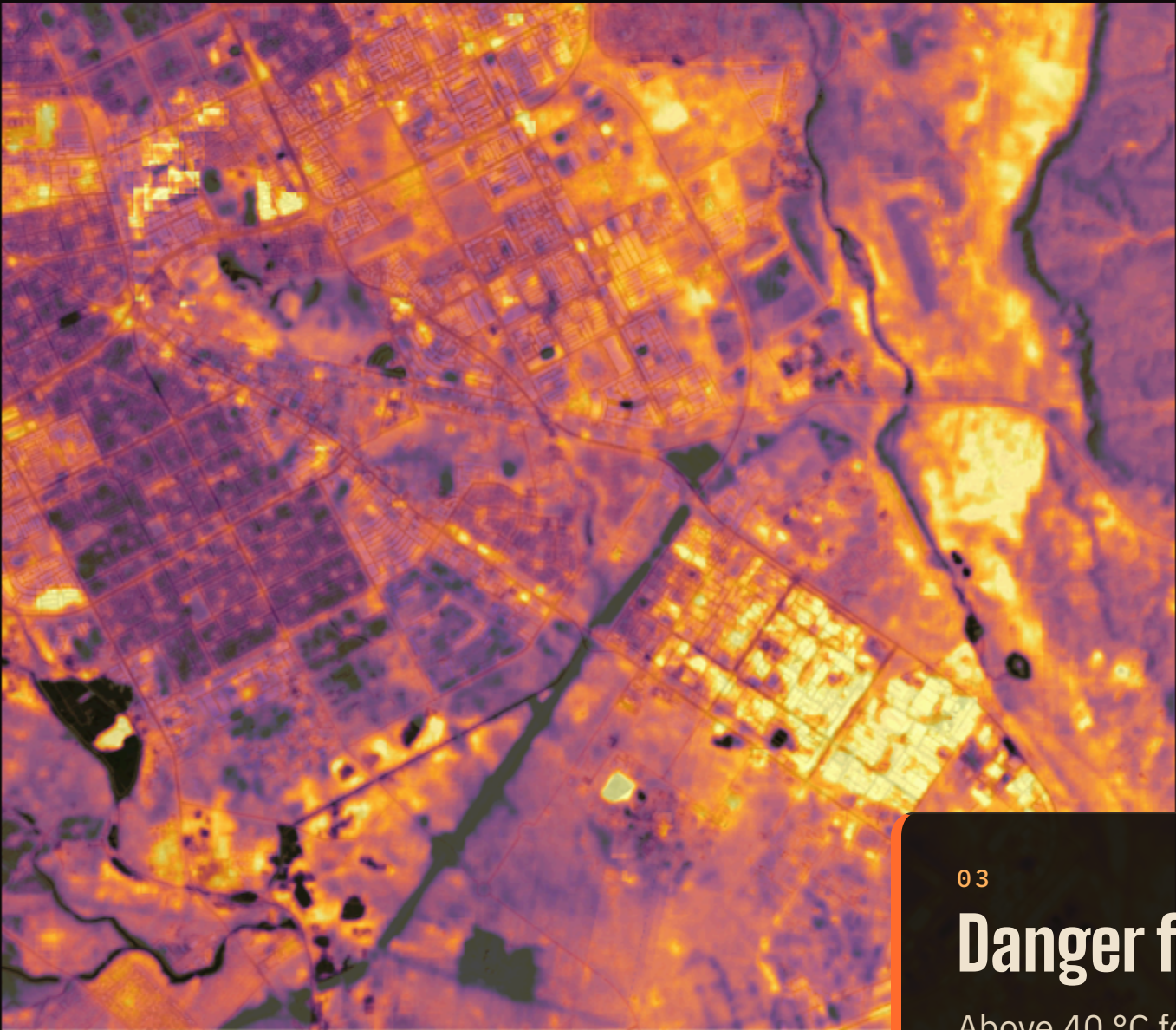
Mosques, bus stops, delivery-rider hubs, construction sites, schools and clinics on WorldPop residents. The hottest fifth of neighbourhoods has 0.00 mapped bus stops per 10,000 people, against 0.63 elsewhere.

157,070

+

WHEN

-



East Riyadh

As Sali · Saudi Arabia

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03

Danger from 11:00

Above 40 °C for 8.3 hours a day, across both afternoon prayers.

04 · WHY

Roof, road or sand? Tanager reads the surface

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0.76

full spectrum F1

0.75

6 broad bands

0.38

starter index rules

05 · WHO

Who is outside in it

Mosques, bus stops, delivery-rider hubs, construction sites, schools and clinics on WorldPop residents. The hottest fifth of neighbourhoods has 0.00 mapped bus stops per 10,000 people, against 0.63 elsewhere.

157,076

residents in hottest 20%

90%

priorities changed by counting people

06 · WHAT NEXT

Where to act first

300 m cells ranked by heat × people × missing greenery. Pick a hotspot to see who is exposed, why it is hot, what to do and the estimated cooling.

-0.70 °C

per +0.10 roof albedo

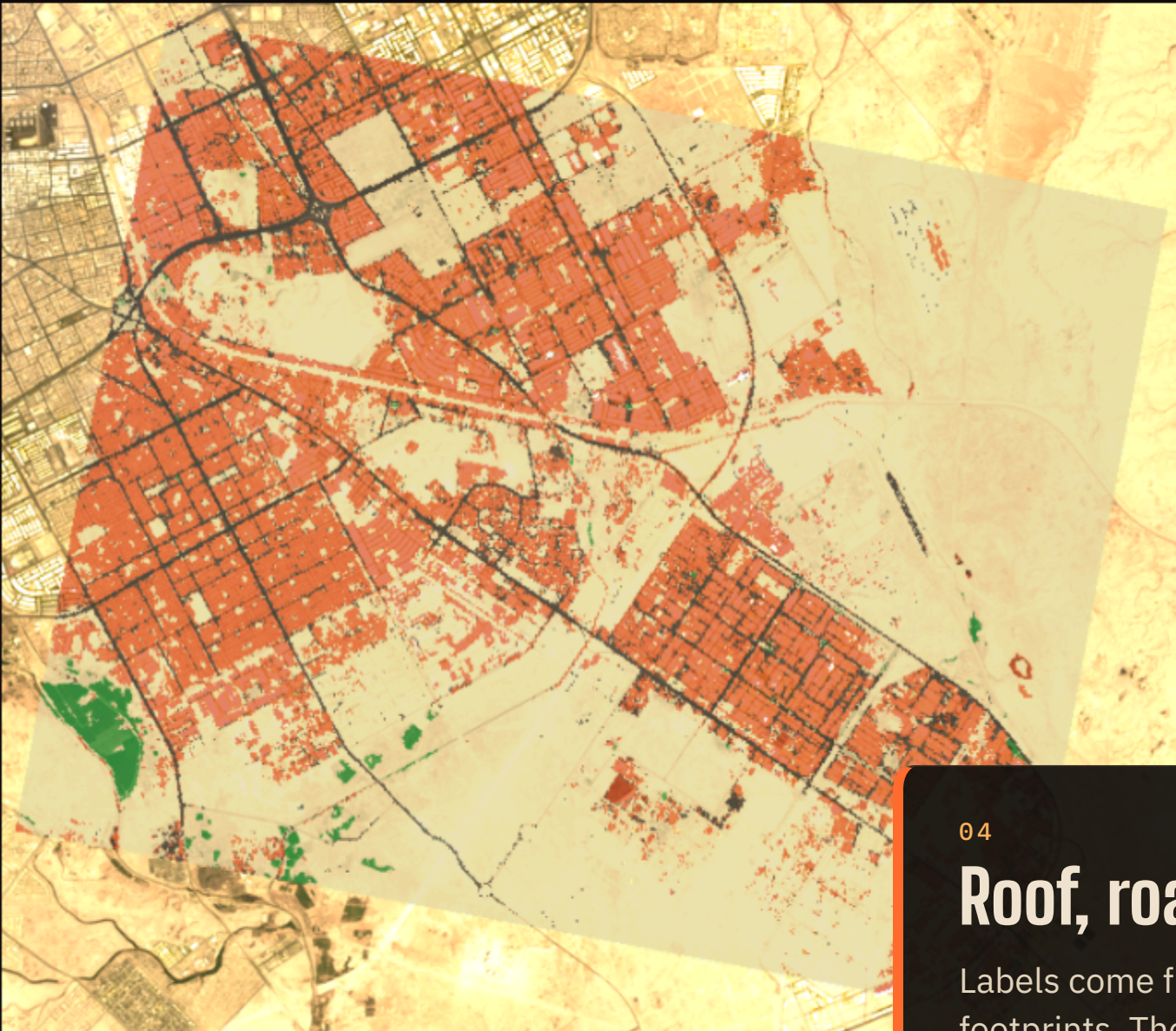
+0.13

R² gained from hyperspectral

+

WHY

-



Surface (Tanager)

- Road / asphalt

Roof / building

Vegetation

Bare / sand

Water

East Riyadh

As Sali · Saudi Arabia

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04

Roof, road or sand

Labels come free from OpenStreetMap and building footprints. The full spectrum scores F1 0.76 on unseen tiles; the starter index rules score 0.38.

Open data: Planet Tanager (CC-BY-4.0), Landsat 8/9, ESA WorldCover, WorldPop 2025, OpenStreetMap, Microsoft building footprints, Open-Meteo/ERA5

05 · WHO

Who is outside in it

Mosques, bus stops, delivery-rider hubs, construction sites, schools and clinics on WorldPop residents. The hottest fifth of neighbourhoods has 0.00 mapped bus stops per 10,000 people, against 0.63 elsewhere.

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-0.70 °C

per +0.10 roof albedo

+0.13

R² gained from hyperspectral

1

403 residents

Cool roofs (reflective coating)

54.4°

2

441 residents

Shaded walkways to mosque

55.7°

3

307 residents

Cool roofs (reflective coating)

56.0°

4

410 residents

Cool roofs (reflective coating)

54.2°

5

282 residents

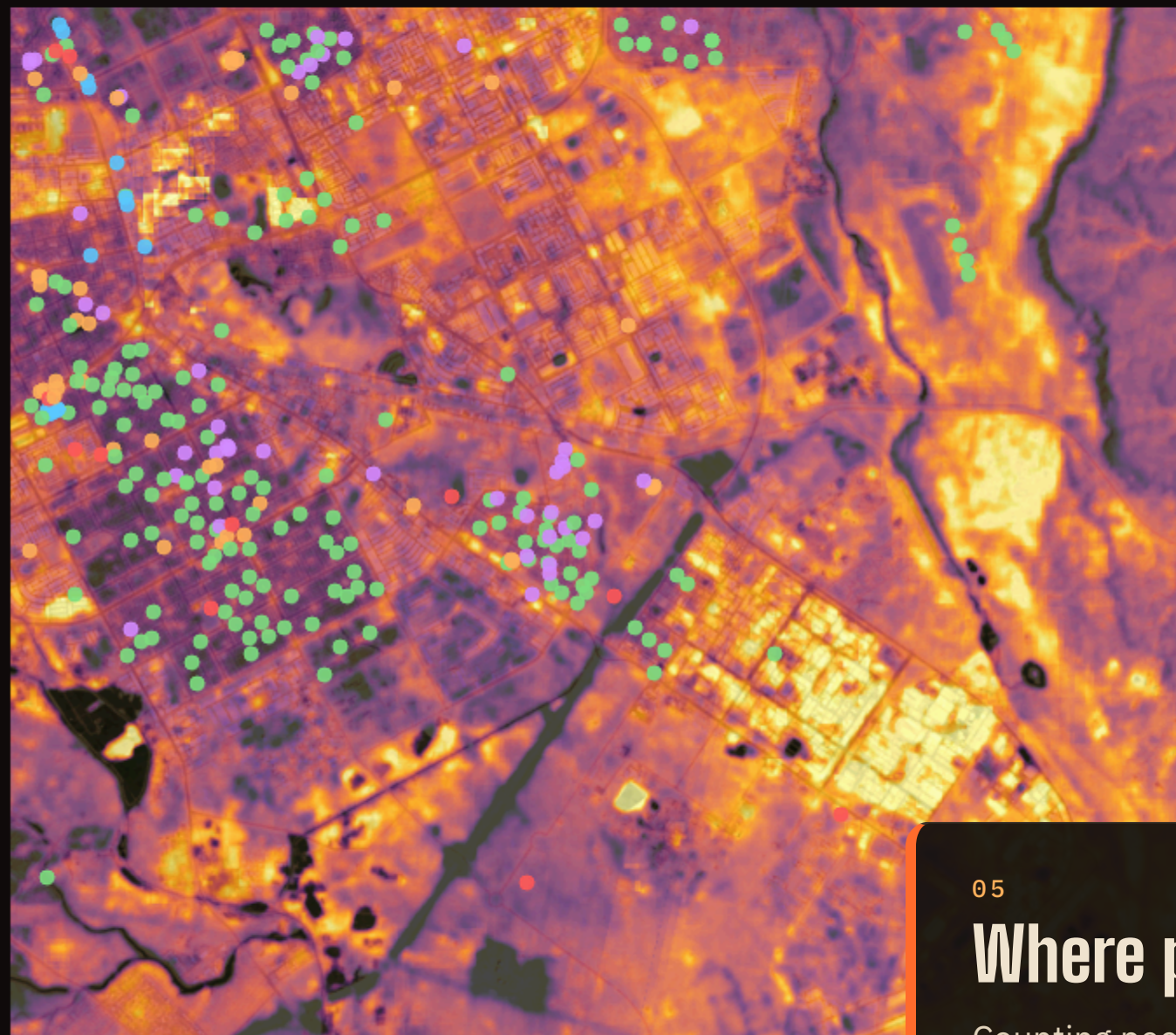
Shaded walkways to mosque

55.9°

WHO

+

-



Surface temperature



● Mosque ● Bus stop ● Rider hub
● Construction ● School / clinic

East Riyadh

As Sali · Saudi Arabia

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05

Where people are outside

Counting people instead of buildings changes 90% of the top-20 priorities.

Open data: Planet Tanager (CC-BY-4.0), Landsat 8/9, ESA WorldCover, WorldPop 2025, OpenStreetMap, Microsoft building footprints, Open-Meteo/ERA5

06 · WHAT NEXT

Where to act first

300 m cells ranked by heat × people × missing greenery. Pick a hotspot to see who is exposed, why it is hot, what to do and the estimated cooling.

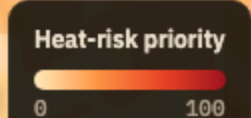
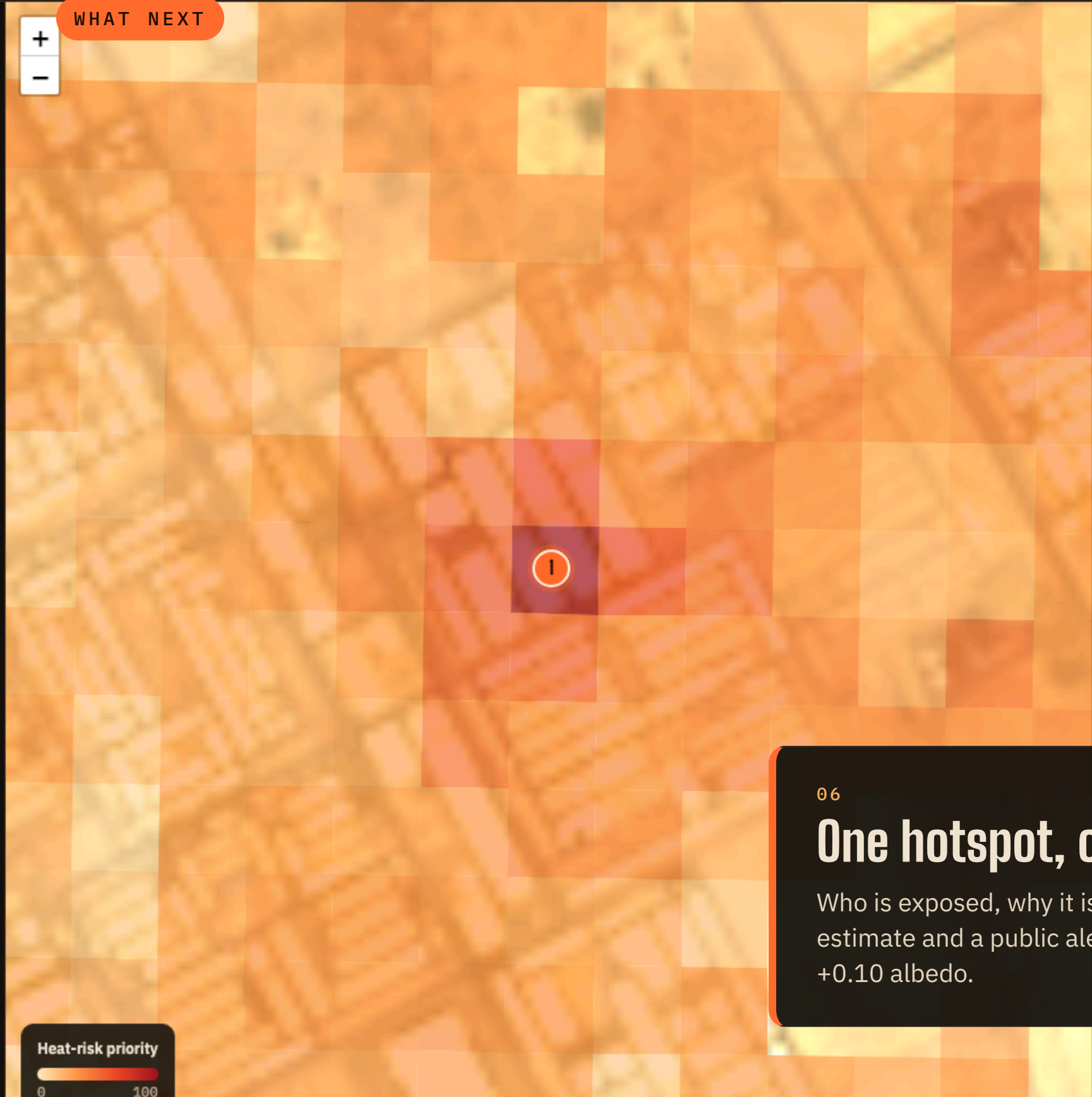
-0.70 °C **+0.13**

per +0.10 roof albedo R² gained from hyperspectral

1	403 residents Cool roofs (reflective coating)	54.4°
2	441 residents Shaded walkways to mosque	55.7°
3	307 residents Cool roofs (reflective coating)	56.0°
4	410 residents Cool roofs (reflective coating)	54.2°
5	282 residents Shaded walkways to mosque	55.9°
6	268 residents Cool roofs (reflective coating)	55.0°
7	527 residents Cool pavement	53.9°
8	449 residents Shaded walkways to mosque	53.5°
9	383 residents Cool roofs (reflective coating)	54.7°
10	402 residents Shade trees / green corridors	54.0°



WHAT NEXT



URGENT

HIGH CONFIDENCE

PRIORITY 100

Hotspot 1

WHERE	24.6220, 46.8640 · open map ↗
HEAT	54.4 °C surface at 10:40 · hot in 3/3 summers
WHO	403 residents
WHY	Mostly roofs (roof 100% · road 0%) · greenery 0%

WHAT NEXT

Cool roofs (reflective coating)

Shade trees / green corridors

Heat-aware design code for new district

WHAT-IF · ESTIMATED ΔLST

Cool roofs -2.80°

Cool pavement +0.00°

Trees +15% +0.00°

Statistical slope with 95% CI, e.g. cool roofs -3.38...-2.26 °C

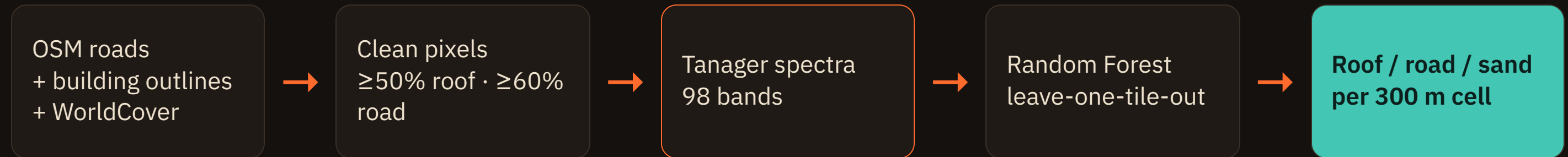
Public alert
Extreme surface heat (54 °C at ~10:40; air ≥40 °C 11:00–20:00). Avoid outdoor waiting, walking and cycling. Use shaded routes, carry water and wear hats and riders.

06

One hotspot, one decision

Who is exposed, why it is hot, what to do, the cooling estimate and a public alert. Cool roofs: -0.70 °C per +0.10 albedo.

Surface intelligence from automatic labels



Why it fits

Spectral shape separates asphalt, roofs and sand that look alike in RGB.

Why Random Forest

Small labels, many bands, explainable. Bigger models did not earn their place.

Why tiles

Testing on unseen tiles stops neighbouring pixels from leaking.

VALIDATION

Every claim has a number

0.89

built-up F1 · spatial CV
starter rule 0.58

0.86

F1 on independent
IO LULC 2023

0.76

surface F1, unseen tiles
6 bands 0.75 · rules 0.38

+0.13

R^2 for heat from adding
hyperspectral (built-up)

r 0.87

Tanager ↔ Landsat
co-registration

60–100%

top-20 stable under
re-weighting

KNOWN TRADEOFFS

Where the edges are

LABELS

Automatic labels carry noise. We keep clear pixels only and test on held-out tiles.

HEAT

Landsat sees surfaces at 10:40. Air peaks +3.2 °C later; weather data sets the hours.

SPECTRA

Full spectrum gains +0.01 F1 over 6 bands for surfaces, +0.13 R^2 for heat. Modest, measured, reported.

PEOPLE

We model where outdoor activity is likely. We do not track people.

02 · WHERE

Masdar reads hottest, and the reason is sand

Masdar City averages 55.4 °C, but only 35% of it is built. The rest is open sand and 8 construction sites. Musaffah: 54.5 °C.

55.4 °C **54.5 °C**
Masdar City Musaffah

03 · WHY

The starter rule calls sand a city

NDBI > 0 labels Musaffah 92% and Masdar 80% built-up. WorldCover says 70% and 17%. QAYDH's model: 82% and 35%.

80% **17%**
NDBI says built (Masdar) WorldCover

04 · WHO

Musaffah: workers, riders, worshippers

Musaffah holds 87,463 residents, 25 bus stops and 36 mosques in one hot industrial grid, the highest people-priority of the four districts.

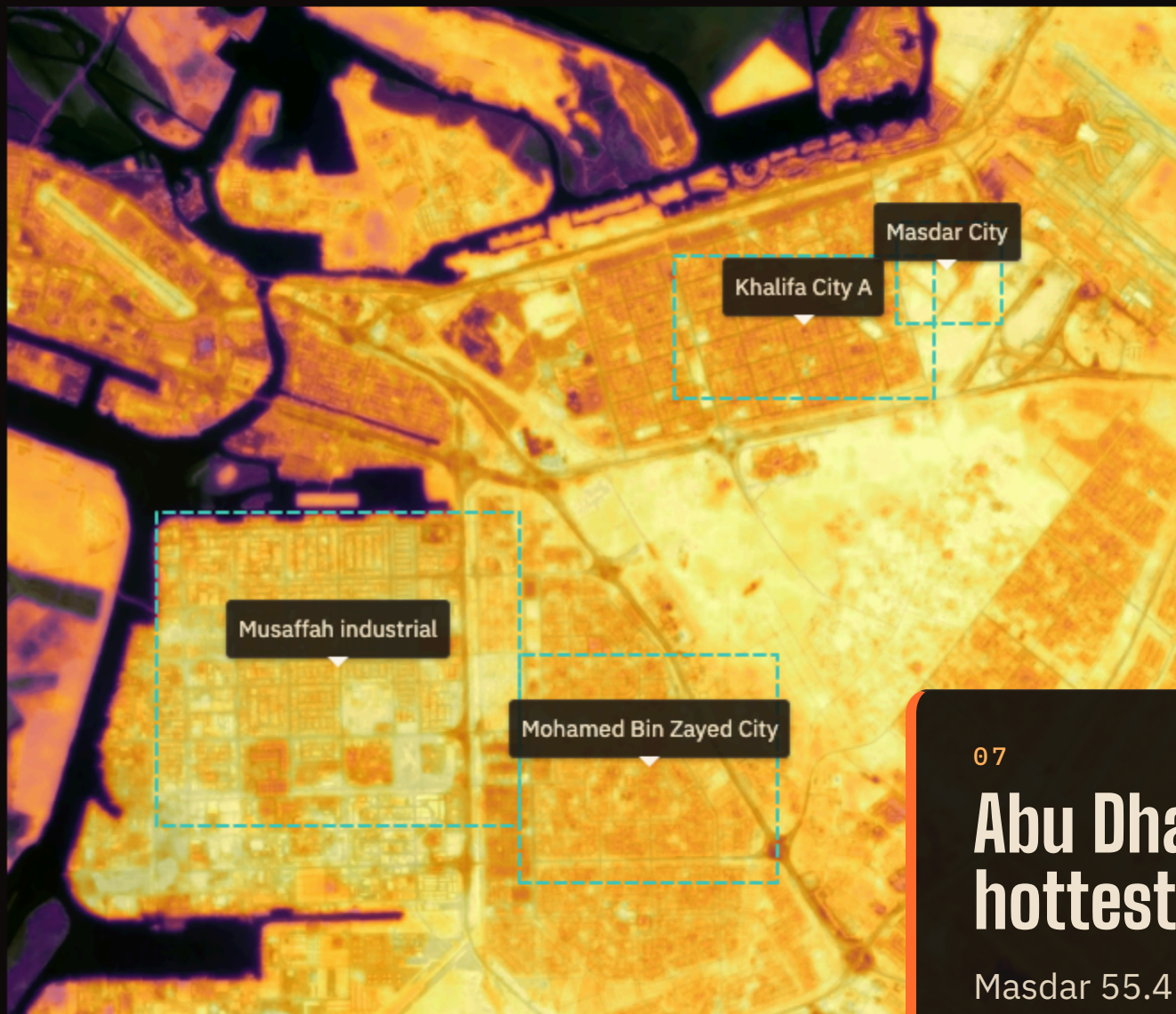
53.7 **43.1**
Musaffah priority Masdar priority

05 · WHAT NEXT

Where to act first in Abu Dhabi

Ranked 300 m cells with the action each one needs. Hyperspectral roof/road detail plugs in when Satellite 813 or MBZ-SAT data is available.

UAE TRANSFER



Surface temperature



Abu Dhabi

Musaffah · Masdar · MBZ City · Khalifa City

WHERE	Persistent summer hotspots at 300 m
WHEN	Hours above 40 °C, from weather reanalysis
WHY	Roof, road, sand or missing greenery, read from hyperspectral spectra
WHO	Residents + people outdoors: mosque-goers, bus riders, delivery riders, workers, children
WHAT NEXT	One named action per hotspot, with estimated °C and confidence

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07

Abu Dhabi: Masdar reads hottest, and the reason is sand

Masdar 55.4 °C with 35% built. The starter rule calls it 80% built and Musaffah 92%. Musaffah ranks first for people: 87,463 residents, 25 bus stops, 36 mosques.

WHO BENEFITS

From map to budget line

Municipalities

Annual city licence: ranked cells, actions, °C estimates, alerts.

Developers & master-planners

Heat-aware design check before districts lock in.

Delivery platforms & contractors

Rest points, shifts and routes for riders and workers.

Dashboard

story map + hotspot cards

API

GeoJSON per 300 m cell

Alerts

summer brief + public heat alerts

INCUBATION

PoC → MVP

DATA

Satellite 813 and MBZ-SAT hyperspectral over Abu Dhabi, Dubai and Al Ain

PEOPLE

Labour housing and delivery-hub layers from municipal and platform partners

TRUTH

Weather-station calibration and 200–300 field-checked surface points

PRODUCT

gIQ dashboard with a what-if slider for roofs, trees and shaded stops

QAYDH القيقظ

Where it burns. Why. Who is outside. What to do first.

[Notebook](#) · [dashboard](#) · [data](#): [GitHub repository](#) [README](#)